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Beyond NL-to-SPARQL: SPO-Based Sentence Generation for Knowledge Graph Question Answering

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ABSTRACT Knowledge graph question answering systems often rely on rule-based natural language to SPARQL translation or sequence-to-sequence generation models, both of which suffer from limited robustness to paraphrasing and complex query structures. This paper proposes a knowledge graph–driven framework that transforms Subject–Predicate–Object (SPO) triples into semantically meaningful natural language sentences and fine-tunes large language models for similarity-based query answering. To enhance semantic coverage, the base SPO sentence corpus is extended through controlled sentence rephrasing, enabling improved alignment between user queries and knowledge graph content. Multiple fine-tuned sentence similarity models are employed and combined using a voting-based ensemble mechanism to mitigate individual model bias and improve retrieval robustness.

The proposed approach is evaluated across numeric, factoid, list, and compound query types using both quantitative and instance-level analyses. Experimental results demonstrate strong regression and ranking performance, with high similarity prediction accuracy and consistent inter-model agreement. Comparative evaluations against rule-based systems such as PAROT and neural machine translation–based SPARQL generation methods further highlight the advantages of the proposed framework in terms of generalization, robustness to paraphrased queries, and ease of interaction. Overall, the results confirm that SPO-centric sentence generation combined with dataset rephrasing and ensemble-based similarity modeling provides a scalable and effective alternative to traditional symbolic and sequence-generation approaches for knowledge graph question answering.

INDEX TERMS Knowledge Graph Question Answering, SPO-Based Sentence Generation, Semantic Similarity Modeling, Large Language Models, Ensemble Learning, Sentence Rephrasing, NL-to-SPARQL Alternatives, SDG 4 (Quality Education), SDG 9 (Industry, Innovation, and Infrastructure).

I. INTRODUCTION

QUESTION ANSWERING (QA) systems have gained significant growth in recent years, serving as pivotal tools for facilitating information retrieval and enhancing user interactions with data [1], [2], [3], [4]. These systems leverage advanced natural language processing techniques to interpret user queries and provide relevant answers, making them invaluable in various fields such as Government services, Entertainment, Healthcare, and beyond [5], with notable advancements in sectors like education [6], [7], [8]. In educational contexts, QA systems empower learners by offering instant access to information, fostering a more engaging and interactive learning experience [9] and administrative efficiency [10], [11] and [12].

The integration of structured knowledge base such as knowledge graphs (KGs) into QA systems represents a transformative approach to information retrieval. When a knowledge graph is used as the underlying knowledge base for answer retrieval, all information must be structured in the form of subject, predicate, and object triples [7], [13], and [14], [15]. Knowledge graphs in QA systems, are employed to retrieve factoid answers, including specific facts, boolean responses, and listbased information, by leveraging the semantic relationships between entities [16], [17], [18], [19], [20] and [21].

Any sentence in its natural language format must be transformed into a structured representation consisting of subject, predicate, and object triples which are the fundamental units of KG in order to retrieve answers from the knowledge

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base effectively. Thus, to retrieve answers for any natural language query, it is necessary to predict the corresponding triples. However, despite this structured approach for answer retrieval, challenges persist in accurately predicting subjects, objects, and predicates [19], [22] and [23]. The task of constructing a comprehensive dataset to train a machine learning or a deep leaning model that accurately predicts these triples and reflects the relationships within the domain of interest remains a significant challenge.

Previous research studies that have been reviewed commonly employ publicly available datasets, such as Freebase, DBpedia, SimpleQuestions, QALD, LC-QuAD, and WebQuestions. However, these datasets often present challenges, particularly when applied to specific domains such as education, where domain-specific relationships may not be adequately captured, and vocabulary mismatches between natural language queries and the structured knowledge base can further hinder accurate answer retrieval [24], [25]. Hence this research work explores an alternative approach to dataset preparation for natural language query (NLQ) to SPARQL conversion. Rather than constructing a dataset specifically to predict the subject, predicate, and object from an NLQ, we propose extracting triples directly from a knowledge graph, isolating the final fragments of the Uniform Resource Identifier (URI) of subject, predicate, and object, and concatenating them to form sentences. These sentences serve as the base dataset, which is further extended by rephrasing them to introduce linguistic variation on which Large Language Models are trained.

The rational behind such a study is to overcome the prediction of Subject (S) , Predicate (P) and Object (0) which requires large, manually-curated datasets designed for SPO prediction, as well as extensive labeling and supervision, which can be time-consuming, domain-specific, and prone to inconsistencies. Additionally, these methods often struggle with the complexity and variability of natural language, leading to inefficiencies in accurately retrieving relevant information.

Based on the challenges investigated and discussed, some potential research questions are framed. They are, RQ1: How effective is the approach of extracting triples from a knowledge graph to form sentences, compared to traditional methods of dataset preparation for predicting SPO of a natural language query? RQ2: How does extending the base dataset through sentence rephrasing influence the performance of large language models (LLMs) in predicting similarity scores? RQ3: How does the proposed method of sentence formation by concatenating subject, predicate, and object triples affect the accuracy of fine-tuned LLMs in retrieving semantically similar sentences?

These research questions can serve as a foundation for investigating the development of a framework which encompasses the construction of a coherent dataset on which LLMs are trained to obtain answer for the user query. The objectives of the proposed work are: (1) To generate a dataset from knowledge graph triples and their rephrased variations using Large Language Models (LLMs). (2) To fine-tune the sentence similarity LLMs on the prepared dataset and showcase the effectiveness of the dataset preparation method. (3) To implement and assess a voting system across six LLMs, evaluating its performance in retrieving similar sentences. (4)To examine the influence of linguistic variation on model performance.

The remainder of this paper is structured as follows. Section II reviews related work in knowledge graph–based question answering, covering symbolic, neural, and large language model–driven approaches. Section III presents the proposed methodology, detailing the domain-specific URI extraction process, SPO-based sentence construction, dataset augmentation, and model fine-tuning strategy. Section IV describes the experimental setup, including evaluation metrics followed by an in-depth analysis of results across numeric, factoid, list, and compound query types. Finally, Section concludes the paper by summarizing the key findings and outlining potential directions for future research.

II. RELATED WORKS

A. STRUCTURED APPROACHES USING SEMANTIC MAPPING TECHNIQUES

Several systems have been developed for translating natural language queries into SPARQL over knowledge graphs, primarily addressing factoid questions. Template-based and triple extraction methods, such as those in [1], [17], utilize linguistic preprocessing and semantic resources (e.g., WordNet) to map query components, and are effective within closed or minimal-domain settings. Interactive and guided systems, including [18], [21], offer user-driven construction or disambiguation of queries, supporting transparency and flexibility, though often at the expense of scalability and automation. Semantic matching and normalization approaches, as in [15], [19], [20], employ string similarity, label matching, and probabilistic models for mapping user queries to KG terms, facilitating robust performance for factoid retrieval, with opportunities for enhancement in mapping accuracy and query diversity. Aggregate and complex query extensions [26], [27] focus on paraphrase dictionaries, graph patterns, or distributional similarity, expanding the scope to filter and aggregate operations. Graph-based methods [13], [28]–[30] use dependency structures or subgraph matching to align queries with KG content, providing reliable results for simple queries and a foundation for more advanced reasoning. Hybrid solutions such as [14] combine SPARQL and keyword search to improve recall in ambiguous scenarios. Grammatical and statistical baselines [31] achieve high precision for basic queries, while recent works [1], [15] highlight the importance of integrating advanced semantic similarity and incremental query construction to further enhance system flexibility and coverage.

B. DEEP LEARNING AND NEURAL APPROACHES FOR SPARQL GENERATION

Early neural models were primarily trained from scratch using parallel datasets of natural language questions and SPARQL queries, with a focus on learning the direct mapping between the two. These approaches include sequence-to-sequence learning with copying mechanisms [32], pointer-generator networks [33], and initial generative models such as SGPT [34]. Building on these foundations, the community explored character-level and sequence-based architectures for predicting subject, object, and predicate triples from queries, using datasets like Freebase and SimpleQuestions for evaluation. This phase saw the rise of models based on GRU [35], CNNs [36], [37], BiLSTM [25], [38], [39], and attention-based neural networks [40], with methods such as conditional focused models [41] and paraphrasing strategies [42] introduced to enhance entity and relation mapping.

Further advancements integrated logistic regression and neural classifiers for improved entity and predicate identification [43]. Despite strong performance on factoid and simple queries—reporting accuracies between 62.7% and 88.3% [35]–[42]—these early approaches were largely limited in their ability to handle compositional, aggregate, or comparative questions, as well as more complex reasoning and multi-hop scenarios [35], [38], [39]. In [44] authors present an extensive evaluation of recent neural approaches for SPARQL query generation from natural language, offering a detailed comparison of models, datasets, and evaluation metrics within the KG-QA landscape.

C. LARGE LANGUAGE MODELS (LLMS) AND PRE-TRAINED TRANSFORMERS FOR SPARQL GENERATION

The most recent advancements [45] involve the use of pre-trained large language models (LLMs) and prompt-based frameworks for SPARQL generation. Approaches such as SPARQLGEN's one-shot prompt-based method [46] and generate-then-retrieve systems like ChatKBQA [47] utilize pre-trained encoders to significantly enhance semantic modeling, robustness to paraphrased input, and generalization across domains, while also reducing reliance on large task-specific datasets. Foundational LLMs such as GPT-3 and its variants [49], as well as recent open-source models like Mistral 7B [50], have further extended the flexibility, adaptability, and performance of SPARQL query generation pipelines, facilitating the handling of a broad spectrum of question types and knowledge graph schemas.

III. METHODOLOGY

A. DOMAIN-SPECIFIC UNIFORM RESOURCE IDENTIFIER RETRIEVAL SYSTEM

Figure 1 illustrates the proposed system for generating a structured dataset of sentence-like constructs from a comprehensive KG. Let the KG be a directed, labeled graph G whose triples $T \subseteq E \times R \times E$ consist of entities E (nodes) and predicates R (edges), each triple (s, p, o) with subject s, predicate p, and object o. In an educational institution setting—where the KG spans multiple domains such as library, sports, student councils, and so on—a user's natural-language question may pertain to any one of these. Once we predict the domain [48] $D \subseteq G$ corresponding to the user's query, we restrict attention to exactly those URIs whose associated domain equals D. Formally, we define the domain-specific subgraph of URIs U_D as:

$$U_D \;=\; \{\, u \mid u \in G,\ \mathrm{domain}(u) = D \}\,. \quad \forall u \in U_D$$

For instance, URIs corresponding to "Library" domain are extracted. Algorithms 1 and 2 describe the procedure for extracting the complete set of URIs from the graph. Algorithm 3 performs domain-based filtering of entity URIs by matching their semantic classes with the domain inferred from the user query [48]. Algorithm 4 facilitates the process by systematically traversing the RDF class hierarchy of an entity URI and identifying inherited superclass memberships required for accurate domain validation. Subsequently, each URI $u \in U_D$ undergoes processing via a "URI Processing Engine," comprising two key operations: the Fragment Extraction Module, extracting meaningful URI fragments $F(u)$, and the Fragment Normalization Module, standardizing these fragments to produce processed fragments $F'(u)$.

Formally, this operation can be represented as:

$$F'(u) = \mathrm{Normalize}(\mathrm{Extract}(u)), \quad \forall u \in U_D$$

As an explicit example, the fragment extraction transforms a URI such as:

- http://www.semanticweb.org/mahe/ontologies/2024/5/untitled-ontology-80/Postgraduate into the fragment "Postgraduate",
- http://www.semanticweb.org/mahe/ontologies/2024/5/untitled-ontology-80/Can_Borrow_Maximum_Four into "Can_Borrow_Maximum_Four".

The Fragment Normalization Module then converts these extracted fragments into a standardized format, for example transforming terms such as "E_Journals" into "E Journals" and "BookBankBooks" into "Book Bank Books". Algorithms 5 and 6 describe the procedure for extracting the fragment of the URI and normalizing the fragment.

The resulting processed URI fragments $F'(u)$ are utilized in constructing coherent sentence-like structures through the "Fragment Sentence Construction Module." This module operates by systematically concatenating fragments representing subjects s, predicates p, and objects o, thus forming a textual representation $S(u)$:

$$S(u) = \mathrm{Concatenate}(F'_s(u), F'_p(u), F'_o(u)), \quad \forall u \in U_D$$

The final dataset $\mathcal{D}$ is obtained by aggregating these constructed sentences across all URIs in the domain:

$$\mathcal{D} = \{S(u) \mid u \in U_D\}$$

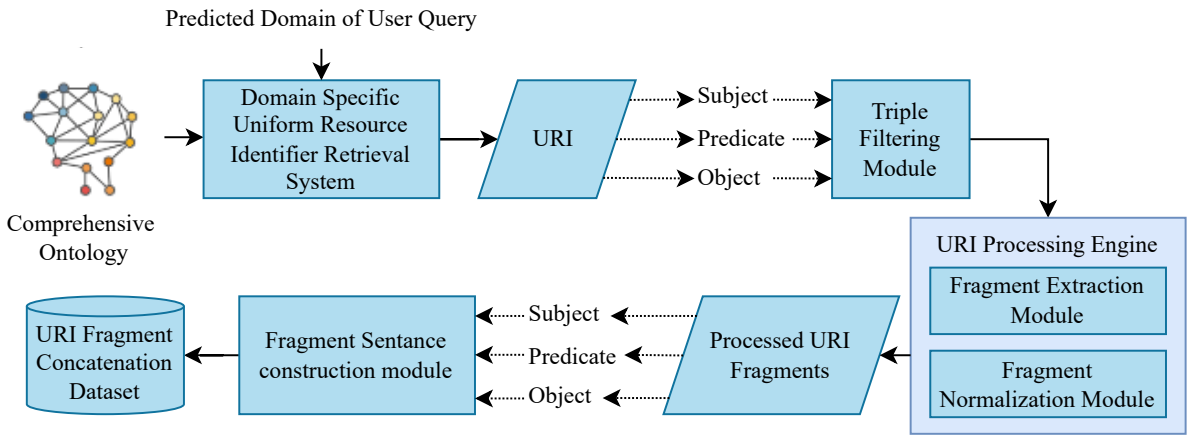

FIGURE 1. **Domain-Specific Uniform Resource Identifier Retrieval System with Sentence Construction from Fragments**

This dataset serves as a foundational corpus for training subsequent neural models. It is important to note that the scope of this paper explicitly excludes detailed examination of domain prediction methodologies and quantitative analyses of the KG structure, referring readers to prior work [48] and future extensions for these aspects.

Algorithm 1 Main Execution

Require: Graph g, predicted domain $domain_input$
Ensure: Matching entities based on the predicted domain

```
entities ← FIND_ALL_ENTITIES(g)
matching_entities ← FILTER_ENTITIES_BY_DOMAIN(entities, domain_input)
return matching_entities
```

Algorithm 2 FIND_ALL_ENTITIES

Require: Graph g
Ensure: Set of all relevant entities

```
Initialize entities ← ∅
for all triples (s, p, o) in g where p = RDF.type do
    if o ∈ {OWL.Class, OWL.NamedIndividual, OWL.DatatypeProperty, OWL.ObjectProperty} then
        entities ← entities ∪ {s}
    end if
end for
return entities
```

B. SYNTHETIC DATASET AUGMENTATION VIA LLAMA FOR ENHANCED LLM FINE-TUNING

Utilizing the pipeline in Figure 1, we augment the corpus via LLaMA and fine-tune five sentence-similarity *encoders—all-mpnet-base-v2*, *stsb-roberta-base*, *paraphrase-MiniLM-L6-v2*, *paraphrase-distilroberta-base-v1*, *distilbert-base-uncased*, and *USE_Lite*.

Algorithm 3 FILTER_ENTITIES_BY_DOMAIN

Require: Set of entities $entities$, target domain $domain$
Ensure: List of entities matching the domain

```
matching_entities ← [ ]
for all entity ∈ entities do
    entity_classes ← FIND_CLASSES_FOR_ENTITY(entity)
    if domain ∈ entity_classes then
        Add (entity, entity_classes) to matching_entities
    end if
end for
return matching_entities
```

Algorithm 4 FIND_CLASSES_FOR_ENTITY

Require: Entity URI $entity_uri$
Ensure: Set of all classes (direct and inherited)

```
classes ← ∅
for all triples (entity_uri, RDF.type, cls) in g do
    Add cls to classes
end for
to_process ← list of elements in classes
while to_process is not empty do
    current ← to_process.pop()
    for all triples (current, RDFS.subClassOf, superclass) in g do
        if superclass ∉ classes then
            Add superclass to classes
            Append superclass to to_process
        end if
    end for
end while
return classes
```

Algorithm 5 Fragment_Extraction_Module

Require: URI uri
Ensure: Last meaningful component of the URI

$parsed \leftarrow PARSE_URI(uri)$
if $parsed.fragment \neq \emptyset$ **then**
  return $parsed.fragment$
end if
if $parsed.path \neq \emptyset$ **then**
  $components \leftarrow SPLIT(parsed.path, '/')$
  for all $component$ in REVERSE($components$) **do**
    if $component \neq \emptyset$ **then**
      return $component$
    end if
  end for
end if
return None

Algorithm 6 Fragment_Normalization_Module

Require: String $text$
Ensure: Normalized version of $text$

if $text$ is not a string **then**
  return $text$
end if
Replace underscores "_" with spaces in $text$
Convert camelCase or PascalCase patterns into space-separated words
Preserve numbers and separate alphabetic–numeric sequences
return $text$

Prompt-Driven Sentence Generation. For each RDF triple (s, p, o) in domain d, we define two domain-specific instruction templates:

- *Positive prompt* π_d^+: Given (s,p,o), write a sentence that faithfully describes this fact with respect to domain d.
- *Negative prompt* π_d^-: Given (s,p,o), write a grammatically plausible sentence that is **not** true of the triple.

C. DOMAIN-ADAPTIVE FINE-TUNING OF SENTENCE-SIMILARITY MODELS FOR EDUCATIONAL QA

In this study, we perform domain-adaptive fine-tuning of several state-of-the-art Sentence-BERT variants to improve semantic similarity estimation specifically for educational question-answering (QA) tasks. The models explored include all-mpnet-base-v2, stsb-roberta-base, paraphrase-MiniLM-L6-v2, paraphrase-distilroberta-base-v1, distilbert-base-uncased, and LaBSE. Each of these models utilizes transformer-based architectures optimized for generating semantically rich sentence embeddings, which are critical for accurately measuring the similarity between natural language queries and candidate answers in educational domains. For hyperparameter tuning on a dataset comprising approximately 20,000 samples, Bayesian optimization [51] presents a highly effective methodology. By utilizing information from previous trials, it strategically navigates the hyperparameter space, concentrating on the most promising areas. This targeted exploration substantially reduces the number of required experiments relative to conventional grid or random search techniques, thereby enhancing computational efficiency and conserving valuable resources. Figure 2 illustrates the fine-tuning process applied to various Sentence-BERT variants. All implementations were performed on an NVIDIA H100 GPU-based system using Python and the Hugging Face Transformers framework. Considering the values in Table 2, although DistilBERT-base-uncased achieves the strongest performance across both regression and ranking metrics, the observed performance variations across models indicate complementary strengths in numerical accuracy and ranking consistency. Therefore, rather than relying on a single model, these results motivate the adoption of a voting-based ensemble strategy that integrates predictions from multiple fine-tuned models

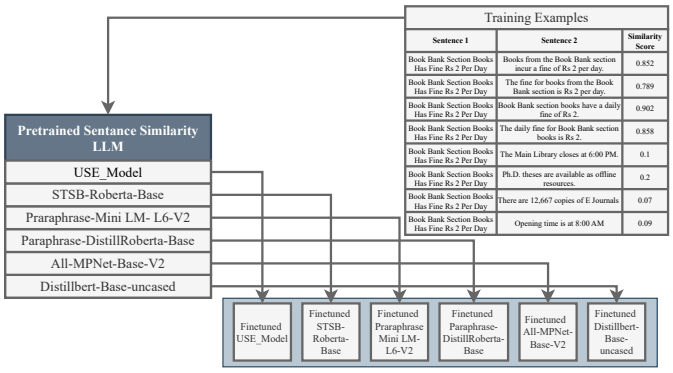

FIGURE 2. **Workflow for Fine-Tuning Pretrained Language Models to Determine Query-Sentence Similarity**

D. TYPES OF QUERIES SUPPORTED BY THE KNOWLEDGE GRAPH

1) Pre-Stored Numeric Value Queries

Figure 3 and Algorithm 7 together illustrate the workflow for handling queries that require retrieval of pre-stored numeric values from the knowledge base. The process begins with query normalization and query-type prediction to identify numeric queries. For such queries, non-numeric candidate sentences are eliminated using pattern-based filtering. The remaining candidate sentences are then evaluated using an ensemble of fine-tuned sentence similarity models, where each model independently computes similarity scores and produces top-ranked results. Finally, a voting-based aggregation mechanism combines model rankings and similarity scores to determine the most relevant numeric answer, thereby improving robustness and reducing model-specific bias.

TABLE 1. Hyperparameter Configurations for Sentence-BERT Variants

Hyperparameter	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Base Model	all-mpnet-base-v2	stsb-roberta-base	paraphrase-MiniLM-L6-v2	paraphrase-distilroberta-base-v1	distilbert-base-uncased	LaBSE
Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW	Adam[1]
Learning Rate	2×10^{-5}	3×10^{-5}	2×10^{-5}	1.5×10^{-5}	2×10^{-5}	2.5×10^{-5}
Batch Size	32	24	32	16	32	24
Number of Epochs	4 (early stopping)	5 (early stopping)	4 (early stopping)	3 (early stopping)	4 (early stopping)	6 (early stopping)
Warm-up Steps	10% total steps	15% total steps	10% total steps	10% total steps	10% total steps	12% total steps
Weight Decay	0.01	0.005	0.01	0.02	0.01	0.005
Early Stopping Patience	3 epochs	4 epochs	3 epochs	3 epochs	5 epochs	3 epochs
Train-Test Split	70%-30%	70%-30%	70%-30%	70%-30%	70%-30%	70%-30%
Loss Function	CosineSimilarityLoss	CosineSimilarityLoss	CosineSimilarityLoss	CosineSimilarityLoss	CosineSimilarityLoss	MSELoss[2]
Scheduler	Linear warmup	Linear warmup	Cosine annealing[3]	Linear warmup	Linear warmup	StepLR scheduler[4]

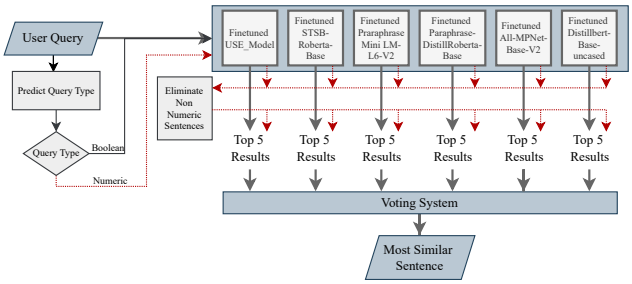

FIGURE 3. Framework for Answering Pre-stored Numeric Answers in Knowledge Graph

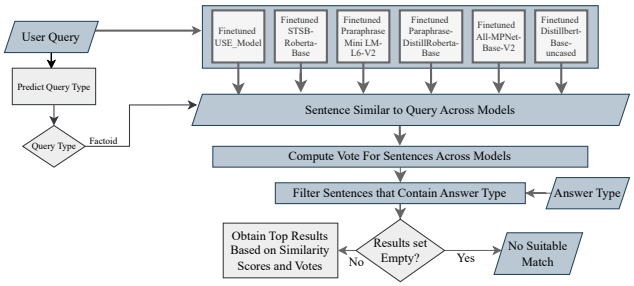

FIGURE 4. Workflow for Answering Factoid type of query

Algorithm 7 Query Processing for Pre-Stored Numeric Answers

Require: User query q, dataset df, set of fine-tuned models $\mathcal{M}$

Ensure: Top ranked sentences answering query q

Load dataset df and fine-tuned models $\mathcal{M}$
Normalize query q by converting to lowercase and removing extra spaces
Initialize similarity scores, vote counts, and combined scores
for each model $m \in \mathcal{M}$ **do**
    Generate embeddings for query q and candidate sentences
    Compute cosine similarity between query and sentence embeddings
    Filter candidate sentences containing numeric values using regular expressions
    Update similarity scores and vote counts
end for
Compute combined score for each sentence using similarity scores and votes
Rank sentences based on the combined score
Select the highest ranked sentence
return Top ranked sentence as final answer

2) Factoid Queries

Figure 4 and Algorithm 8 illustrate the workflow for answering factoid-type queries using an ensemble of fine-tuned sentence similarity models. After an initial preprocessing stage and query classified as factoid, the query is then evaluated across multiple fine-tuned models to compute sentence-level similarity scores. Candidate sentences are aggregated through a voting-based mechanism that combines similarity rankings across models. To ensure semantic correctness, the aggregated results are further filtered based on the expected answer type. Finally, the top-ranked sentences are selected as factoid answers, while cases with no suitable matches are explicitly handled.

3) List-Type Queries

Figure 5, together with Algorithm 9, illustrates the workflow for answering list-type queries using an ensemble of fine-tuned sentence similarity models. The user query is first classified to identify the list query type and evaluated across multiple models to compute sentence-level similarity scores. Candidate sentences are aggregated through a voting-based mechanism and filtered based on the expected answer type. The final output consists of a ranked list of relevant sentences, while cases with no suitable matches are explicitly handled.

4) Compound Queries

The Figure 6 and Algorithm 11 describes the procedure for answering compound queries by decomposing them into simpler sub-queries shown in Algorithm 10 and aggregating results using an ensemble-based voting mechanism. The system first analyzes the sentence structure to identify compound statements, which are then decomposed into multiple simpler sub-queries using rule-based parsing. Each simplified query is independently processed using an ensemble of fine-tuned sentence similarity models. The resulting

Algorithm 8 Answering Factoid-Type Queries

Require: User query q, dataset df, set of fine-tuned models $\mathcal{M}$, expected answer type T
Ensure: Top ranked sentences answering query q
Load dataset df and fine-tuned models $\mathcal{M}$
Normalize query q by converting to lowercase and removing extra spaces
Initialize vote counts, similarity scores, and combined scores
for each model $m \in \mathcal{M}$ **do**
    Generate embeddings for query q and candidate sentences
    Compute cosine similarity between query and sentence embeddings
    Update similarity scores and vote counts
end for
Compute combined score for each sentence using votes and similarity scores
Store ranked candidates in C
Initialize an empty list $C_{filtered}$
for each sentence $s \in C$ **do**
    if $contains_entity(s, T)$ is **true then**
        Add s to $C_{filtered}$
    end if
end for
Rank $C_{filtered}$ based on combined scores
Select top ranked sentences
return Top ranked sentences as final answers

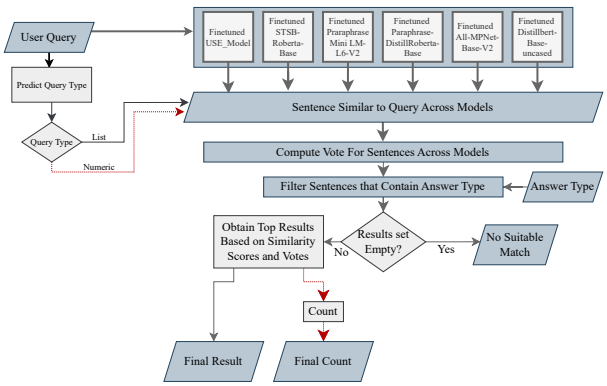

FIGURE 5. **Workflow for answering list-type queries**

candidate sentences are aggregated through a voting-based mechanism and filtered based on the expected answer type. Finally, duplicate results are removed and logical operators such as and, or, and but not are applied to combine the intermediate results, producing a coherent and semantically consistent final answer set.

IV. RESULTS & ANALYSIS

The Table 2 presents the performance comparison of various fine-tuned text similarity models using regression-based and rank-based evaluation metrics, namely MSE, MAE,

Algorithm 9 Answering List-Type Queries

Require: User query q, dataset df, set of fine-tuned models $\mathcal{M}$, expected answer type T
Ensure: Ranked list of sentences answering query q
Load dataset df and fine-tuned models $\mathcal{M}$
Normalize query q by converting to lowercase and removing extra spaces
Initialize vote counts, similarity scores, and combined scores
for each model $m \in \mathcal{M}$ **do**
    Generate embeddings for query q and candidate sentences
    Compute cosine similarity between query and sentence embeddings
    Update similarity scores and vote counts
end for
Compute combined score for each sentence using votes and similarity scores
Store ranked candidates in C
Initialize empty list $C_{filtered}$
for each sentence $s \in C$ **do**
    if $contains_entity(s, T)$ is **true then**
        Add s to $C_{filtered}$
    end if
end for
if $C_{filtered}$ is empty **then**
    return No suitable sentences found
end if
Rank $C_{filtered}$ based on combined scores
return Top-N ranked sentences as list answers

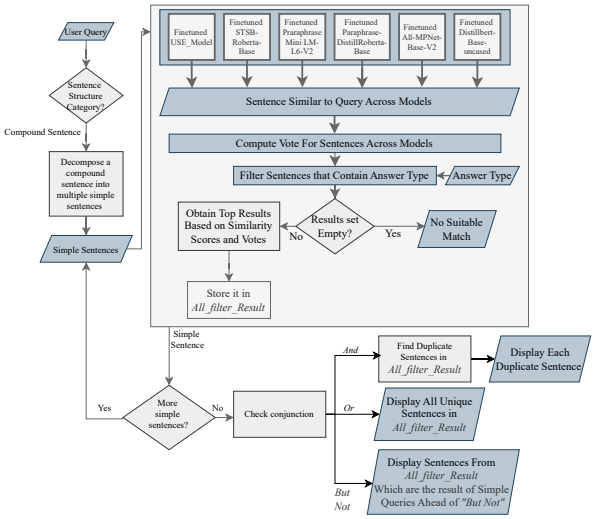

FIGURE 6. **Workflow to Answer Compound Queries**

R^2 score, Spearman's rank correlation, and Kendall's Tau. From regression performance perspective, among all models, DistilBERT-base-uncased achieves the lowest MSE (0.0031) and lowest MAE (0.0360), indicating the most accurate

Algorithm 10 Decomposition of Compound Statements into Simpler Sentences

Require: Compound sentence S
Ensure: Set of decomposed simpler sentences

Extract *base_query* and *categories_part* from S using regular expressions
  base_query ← text occurring before the phrase "which are"
  categories_part ← text occurring after the phrase "which are"
Initialize empty lists: *included_categories* and *excluded_categories*
if "but not" occurs in *categories_part* **then**
    Split *categories_part* into two segments around "but not"
    included_categories ← categories before "but not" (split using "and" / "or")
    excluded_categories ← category following "but not"
else
    included_categories ← split *categories_part* using "and" / "or"
end if
Generate simpler sentences by combining *base_query* with each category in *included_categories*
Exclude sentences corresponding to categories in *excluded_categories*
return set of valid decomposed sentences

Algorithm 11 Answering Compound Queries

Require: User query q, dataset df, set of fine-tuned models $\mathcal{M}$, expected answer type T
Ensure: Set of sentences answering compound query q

Load dataset df and fine-tuned models $\mathcal{M}$
Normalize query q by converting to lowercase and removing extra spaces
Decompose compound query q into simpler sub-queries using Algorithm 10
Store decomposed queries in set $\mathcal{Q}_{simple}$
Initialize empty set A_{all} to store filtered answers
for each simple query $q_i \in \mathcal{Q}_{simple}$ **do**
    Initialize vote counts, similarity scores, and combined scores
    for each model $m \in \mathcal{M}$ **do**
        Generate embeddings for q_i and candidate sentences
        Compute cosine similarity between query and sentence embeddings
        Update similarity scores and vote counts
    end for
    Filter candidate sentences based on expected answer type T
    Add filtered results to A_{all}
end for
Aggregate results in A_{all} using logical operators (AND / OR / BUT NOT)
Remove duplicate sentences
Rank final sentences based on combined similarity scores
return ranked sentences as final answers

numerical prediction of similarity scores. It also records the highest R² score (0.9607), suggesting superior variance explanation and an excellent model fit. MPNet follows closely, with competitive MSE (0.0042) and MAE (0.0420), and a strong R² score of 0.9465, confirming its robustness in capturing semantic similarity. In contrast, USE-base exhibits significantly higher error values (MSE = 0.0183, MAE = 0.0900) and the lowest R² score (0.7735), indicating comparatively weaker regression performance. From a ranking perspective, DistilBERT-base-uncased again demonstrates the strongest performance, achieving the highest Spearman's rank correlation (0.9329) and Kendall's Tau (0.8013). This highlights its ability to preserve the relative ordering of sentence similarity pairs effectively. MPNet also shows strong rank correlation (Spearman = 0.9226, Kendall = 0.7806), making it a reliable alternative where both regression accuracy and ranking consistency are required. The paraphrase-distilRoBERTa-base-v1 model performs moderately well in ranking metrics but lags behind in regression accuracy, suggesting a trade-off between numerical precision and ordering capability. In summary, DistilBERT-base-uncased emerges as the best-performing model across both regression and ranking metrics.

TABLE 2. Performance measures for various text similarity models

Model	MSE	MAE	R^2	Spearman	Kendall
MPNet	0.0042	0.0420	0.9465	0.9226	0.7806
STS-B RoBERTa	0.0052	0.0552	0.9342	0.8733	0.7070
paraphrase-MiniLM	0.0060	0.0497	0.9236	0.8642	0.7082
paraphrase-distilRoBERTa-base-v1	0.0064	0.0510	0.9191	0.8948	0.7398
DistilBERT-base-uncased	**0.0031**	**0.0360**	**0.9607**	**0.9329**	**0.8013**
USE-base	0.0183	0.0900	0.7735	0.8022	0.6190

A. INSTANCE-LEVEL ANALYSIS OF QUERY ANSWERING

Table 3 presents representative instances of pre-stored numeric queries along with the top-voted answer sentences and similarity scores obtained from individual fine-tuned models. The number of votes reflects inter-model agreement, while the total and combined similarity scores indicate aggregated confidence. Despite variations in similarity scores across individual models, the voting-based aggregation consistently promotes the correct numeric sentence, demonstrating robustness in retrieving pre-stored numerical facts from the knowledge graph. Table 4 presents the top-voted sentences selected for factoid-type queries along with their similarity scores computed using multiple language models. Table 5 presents the results for the list type of query "Who are the working staff in the library?" and Table 6 shows the results

TABLE 3. Top Voted Sentences and their Similarity Scores for queries on Pre-stored Numeric Answers in Knowledge Graph

Query	Top Voted Sentence	Number of Votes	Similarity Score from Models						Total Similarity	Combined Similarity
			all-mpnet-base-v2	stsb-roberta-base	paraphrase-MiniLM-L6-v2	paraphrase-distilroberta-base-v1	distilbert-base-uncased	USE_Lite		
How many E Journals are there?	Online Journals Number Of Copies 2750	6	0.3421	0.5557	0.3987	0.2456	0.3351	0.2456	2.1228	8.1228
For how many days books can be borrowed from general books section?	General Books Can Be Borrowed For Maximum 14 Days	5	0.6929	0.6505	0.8689	0.65	0.555	0.545	3.9623	8.9623
What is the fine for late submission of books in Text book Section?	Text Book Section Books Has Fine Rs 25 Per Day	4	0.3479	0.5363	0.4205	0.2919	0.2312	0.2456	2.0734	6.0734
What is the seating capacity in the library?	Seating Capacity 950	4	0.4212	0.2175	0.492	0.3451	0.4523	0.3413	2.2694	6.2694

TABLE 4. Top Voted Sentences and their Similarity Scores for Factoid type of queries

Query	Top Voted Sentence	Number of Votes	Similarity Score from Models						Total Similarity	Combined Similarity
			all-mpnet-base-v2	stsb-roberta-base	paraphrase-MiniLM-L6-v2	paraphrase-distilroberta-base-v1	distilbert-base-uncased	USE_Lite		
Who is the chief librarian?	Andrew Walker type Chief Librarian	5	0.5122	0.6433	0.5488	0.4551	0.3451	0.3456	2.5045	7.5045
Whom does Nagesh Prabhu report to?	Andrew Walker Works Under Kevin Peterson	5	0.7653	0.6916	0.645	0.4649	0.7453	0.5678	3.8799	8.8799
What is the genre of The Diary of a Young Girl book?	The Diary of a Young Girl genre Biography	5	0.7844	0.7897	0.8847	0.6326	0.8189	0.6574	4.5677	9.5677
How are project reports available?	Project Reports type Offline Resources	5	0.6152	0.3839	0.671	0.6419	0.6092	0.6789	3.6001	8.6001
For how long books from book bank section be borrowed?	Book Bank Books Can Be Borrowed For Maximum 1 Semester	5	0.4365	0.6335	0.6237	0.5738	0.6091	0.6789	3.555	8.555

for the query "Which are the various books available in the library?" . Table 7 illustrates the compound query decomposition into simpler queries. The Figure 7 illustrates the result of processing a compound query: "Which are the books with Thriller and Fiction genre." The query is decomposed into two sub-queries corresponding to the individual genres Thriller and Fiction. Each sub-query retrieves a ranked list of candidate answers with associated similarity scores. The intersection of the results from both sub-queries identifies "The Shining" as the common book satisfying both genre constraints, demonstrating the effectiveness of the compound query resolution mechanism.

TABLE 5. Who are the working staff in the library?

Query	Sentence	Votes	Similarity Score
Who are the working staff in the library?	John Anderson	5	0.6927
	Emily Carter	5	0.6009
	Michael Thompson	5	0.6858
	Sophia Williams	5	0.6721
	Christopher Brown	5	0.6243
	Daniel Rodriguez	5	0.6205
	Olivia Martinez	5	0.6416
	Laura Mitchell	5	0.6132
	Dr. Rebecca D. Parker	5	0.6196

TABLE 6. Which are the various books available in the library?

Query	Sentence	Votes	Similarity Score
Which are the various books available in the library?	A Brief History of Time	5	0.7395
	The Catcher in the Rye	5	0.6955
	The Great Gatsby	5	0.7126
	To Kill a Mockingbird	5	0.6717
	The Diary of a Young Girl	5	0.4733
	Steve Jobs	5	0.3564

B. COMPARATIVE ANALYSIS

PAROT [52] is a rule-based natural language to SPARQL translation system that relies on ontology-specific lexicons, static embeddings, and handcrafted dependency rules to generate ASK/SELECT queries. While effective for structured and known query patterns, it is limited in handling paraphrases, compound queries, and unseen natural-language formulations [52]. Quantitative results presented in Table 8 further demonstrate the effectiveness of the proposed approach. Our method achieves an accuracy of 0.7843 and an F-measure of 0.63, substantially outperforming PAROT, which records uniform scores of 0.4509 across accuracy, precision, recall, and F-measure. The improvement in recall (0.78 vs. 0.4509) highlights the proposed system's superior ability to retrieve relevant answers, while the higher precision reflects reduced dependence on rigid symbolic matching. Overall, these results indicate that the proposed approach offers better robustness, improved generalization to unseen paraphrases, and enhanced user experience compared to PAROT, albeit at the cost of higher computational complexity due to transformer-based processing.

To further evaluate the proposed framework, we compare it with a neural machine translation–based approach for SPARQL query generation using the T5 model, as discussed by [44]. In this formulation, natural language questions are treated as a text-to-text generation task, where each input query is directly mapped to a target SPARQL query. The model is trained on 150 paired samples using maximum likelihood estimation, minimizing the negative log-likelihood loss. As reported in Table 9, the model demonstrates stable convergence with a training loss of 0.0124 and a validation loss of 0.0751 at the final epoch. While high BLEU (76.05) and F1 (0.7636) scores indicate strong sequence-

TABLE 7. Illustration of compound query decomposition into simpler queries

Original Compound Query	Decomposed Simple Queries
Which are the books which are biography but not humorous?	• Which are the books which are biography? • Which are the books which are not humorous?
Which are the books which are furious and humorous?	• Which are the books which are furious? • Which are the books which are humorous?
Find books which are fiction or mystery but not horror	• Find books which are fiction • Find books which are mystery • Find books which are not horror
List students which are physics but not chemistry?	• List students which are physics • List students which are not chemistry
Which are the books which are thriller and drama but not fantasy?	• Which are the books which are thriller? • Which are the books which are drama? • Which are the books which are not fantasy?
Which are the books which are history?	• Which are the books which are history?

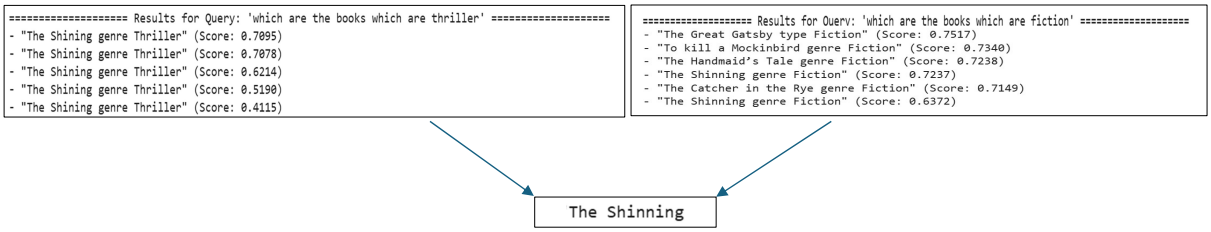

FIGURE 7. The result of the compound query "Which are the books with Thriller and Fiction genre"

TABLE 8. Quantitative comparison of PAROT and the proposed approach

Method	Accuracy	Precision	Recall	F-measure
Proposed Approach	0.7843	0.53	0.78	0.63
PAROT [52]	0.4509	0.4509	0.4509	0.4509

TABLE 9. Performance measures of the T5-based neural machine translation model for SPARQL query generation at the final epoch

Epoch	Training Loss	Validation Loss	BLEU	F1	Relaxed Accuracy
20	0.0124	0.0751	76.05	0.7636	0.3636

level alignment between generated and reference SPARQL queries, the relatively lower relaxed accuracy (0.3636) reveals challenges in achieving exact or structurally consistent query generation for complex and paraphrased inputs. These results highlight the strengths of NMT-based approaches in syntactic mapping, while motivating the need for the proposed semantic retrieval and ensemble-based framework to improve robustness and generalization.

C. RESEARCH QUESTION ANALYSIS

Following the analysis of the results, we now focus our attention to addressing the primary research questions posed at the beginning of this study.

RQ1: How effective is extracting triples from a knowledge graph to form sentences, compared to traditional dataset preparation methods for predicting SPO? The results demonstrate that forming training and retrieval sentences directly from knowledge graph triples is an effective alternative to traditional dataset preparation methods that rely on ontology-specific lexicons and handcrafted rules. Compared to symbolic approaches such as PAROT, which require manual synonym construction and strict ontology-aware phrasing, the proposed SPO-based sentence formation enables natural-language alignment without explicit SPARQL template generation. This is evidenced by the substantial improvement over PAROT in quantitative performance, with accuracy increasing from 0.4509 to 0.7843 and F-measure from 0.4509 to 0.63. Additionally, qualitative analysis shows that the SPO-generated sentences generalize well to paraphrased and compound queries, indicating that KG-driven sentence construction provides a scalable and semantically robust foundation for SPO prediction tasks.

RQ2: How does extending the base dataset through sentence rephrasing influence the performance of LLMs in predicting similarity scores? Extending the base dataset through sentence rephrasing significantly improves the ability of fine-tuned LLMs to predict semantic similarity scores consistently across diverse query formulations. Experimental results across multiple fine-tuned sentence similarity models (e.g., DistilBERT, MPNet, and RoBERTa variants) demonstrate strong regression and ranking performance, with R^2 scores exceeding 0.94 and Spearman's rank correlations above 0.92 for the best-performing configurations. Instance-level analysis further confirms that rephrased SPO sentences enable models to assign high similarity scores even when surface lexical forms vary. Overall, these findings indicate that sentence rephrasing enhances semantic coverage, reduces sensitivity to exact wording, and leads to more stable similarity estimation across models.

RQ3: How does SPO-based sentence formation affect the accuracy of fine-tuned LLMs in retrieving semantically similar sentences? The SPO-based sentence formation strategy positively impacts the accuracy of fine-tuned LLMs in retrieving semantically similar sentences by enforcing a structured yet natural representation of knowledge graph facts. Results from numeric, factoid, list, and compound query evaluations show that sentences derived from SPO concatenation are consistently retrieved as top-ranked candidates across models, with strong inter-model agreement reflected through voting-based aggregation. The ensemble framework further enhances retrieval accuracy by

mitigating individual model biases, as observed in high combined similarity scores and correct answer selection even when individual model scores vary. Overall, the SPO-centric sentence representation enables fine-tuned LLMs to better capture relational semantics, leading to reliable retrieval performance across simple and complex query types.

V. CONCLUSION AND FUTURE WORKS

This work presented a knowledge graph–driven framework for natural language query answering by transforming subject–predicate–object (SPO) triples into semantically meaningful sentences and leveraging fine-tuned large language models for similarity-based retrieval. By systematically generating SPO-based sentences and extending the dataset through controlled rephrasing, the proposed approach enables robust semantic alignment between user queries and knowledge graph content without requiring explicit SPARQL query generation. Experimental results across numeric, factoid, list, and compound query types demonstrate that the ensemble-based voting mechanism effectively exploits complementary strengths of multiple fine-tuned models, leading to consistent retrieval performance even when individual model scores vary.

Comparative evaluations against rule-based systems such as PAROT and neural machine translation–based SPARQL generation approaches further highlight the advantages of the proposed method in terms of robustness, generalization to paraphrased queries, and ease of interaction. Overall, the results confirm that SPO-centric sentence formation combined with dataset rephrasing and ensemble similarity modeling offers a scalable and semantically grounded alternative to traditional symbolic and sequence-generation methods for knowledge graph question answering.

While the proposed framework demonstrates strong performance on static knowledge graph snapshots, real-world knowledge graphs are inherently dynamic, with frequent updates in entities, relationships, and attribute values. As future work, the framework can be extended to support incremental updates by automatically regenerating SPO-based sentences only for modified or newly added triples, rather than rebuilding the entire sentence corpus. Such an approach would reduce computational overhead and ensure timely synchronization between the knowledge graph and the sentence repository.

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